



AI of future

Artificial Intelligence is going to shape the future, but how? Demis Hassabis explains it better than we could: <http://dgit.in/DeepAI>



Nasa's experiment

Nasa as another early adopter of 3D started experimenting with Vive VR headset by letting people experience ISS: <http://dgit.in/ViveVRExp>



Mir as seen on 12th June 1998

cosmonaut along with a support crew to the station in July in order to preserve Mir's permanent manned outpost status.

Despite their best efforts, Mir could not sustain its permanent expedition status, and had to be eventually abandoned for a few months in 1989 when ground testing damaged a Soyuz spacecraft headed to the station. This incident proved to be only a small blip in the record, as Mir would go on to remain occupied for near a decade.

ISS and the new age of cooperation

Despite the success of the Mir space station, there were numerous reports emerging about the apparent and imminent end of the Soviet space program owing to the collapse of the Soviet Union. These rumors were squashed after the creation of the Russian Space Agency, the leaders of which quickly moved to make an agreement of cooperation with NASA for a joint space venture. The likes of such a venture were not thought of since the conception of the Apollo-Soyuz Test Project in 1972, a space program that marked first joint U.S.-Soviet space flight as well as the last flight of an Apollo Program spacecraft.

The first of these new cooperative programs was the Shuttle-Mir program, which saw a total of nine American astronauts visit Mir between 1995 and 1998. These missions gave NASA astronauts their first exposure to long-duration space-flight since Skylab, while teaching both nations the importance of cooperating towards building a new and more-improved modular manned space station. For the betterment of space travel, these plans

did eventually come to fruition and marked the birth of a space program that would bring to life to the now world-famous International Space Station (ISS).

Similar to the Mir, the ISS is an low Earth orbit man-made satellite and is currently the largest artificial body in orbit (which, in the right circumstances, can be seen with the naked eye). True to the program's spirit of international cooperation, early modules for the ISS were launched by Russian Soyuz rockets as well as American Space Shuttles (until the latter program was shut down in 2011). In fact, the ISS program is a joint venture that includes not just NASA and Roscosmos (Russian Federal Space Agency) but also JAXA (Japan Aerospace eXploration Agency), CSA (Canadian Space Agency) and ESA (European Space Agency), which includes 22 member nations.

One of the main objectives of the ISS is to provides a unique platform for conducting scientific research, chief of these being microgravity conditions and exposure to the harsh, unforgiving conditions of space. In addition, space stations like ISS offer an environment in which studies can be performed for potentially decades, something that was not previously possible with manned spacecraft. This opportunity has paved the way for extremely interesting experiments from developing technology to studies on the effect of space-like conditions on human health.

COOL SPACE EXPERIMENT

The effect of boiling water in space is a unique phenomenon that was first observed by astronauts aboard the ISS. On earth, boiling water creates tiny bubbles that rise from the bottom of the vessel to the top, effectively transferring heat across the water being boiled. In space, however, hot air cannot rise since there is no gravity to force the colder air to the bottom. As a result, the bubble of heat formed by boiling the water in space just continues to grow trapping all the heat within it and ensuring the boiling water does not even get warm.

Reports on 28th March 2015 claimed an agreement had be orchestrated between Roscosmos and NASA to collaborate on the development of a replacement for the ISS. This report was based primarily on an announcement by Igor Komarov, the head of Russia's Roscosmos, who stated that "Roscosmos together with NASA will work on the programme of a future orbital station". He also stated that the first step towards this would be to continue operations on the ISS up to 2024. While this sounds like exciting news, NASA spokesman David



SpaceX Dragon capsule on its way to deliver cargo to the ISS

Weaver stated that the agency appreciated the Russian commitment to extending the ISS but he did not confirm any plans for a future space station.

Bringing the final frontier closer to home

Space stations like the ISS also serve as a destination for newer spacecrafts designed by space agencies including national bodies like NASA and JAXA as well as private-owned ones like SpaceX. Since it's launch, the ISS has welcomed a variety of space vessels including prominent ones like NASA's Space Shuttle, Russia's Soyuz, Japan's HTV (H-II Transfer Vehicle), the ESA's ATV (Automated Transfer Vehicle), and most recently SpaceX's Dragon capsule. We haven't even mentioned the potential applications for space tourism, highlighted by Boeing and Sierra Nevada Corporation's plans to dock their manned CST-100 capsule and Dream Chaser space plane, respectively, with the ISS. With the increased interest in space travel and the tangible advances in space exploration technology, it seems only a matter of time before feasible commercial space tourism becomes a reality.